



IoT Use Case Slides

AWS Academy Data Engineering

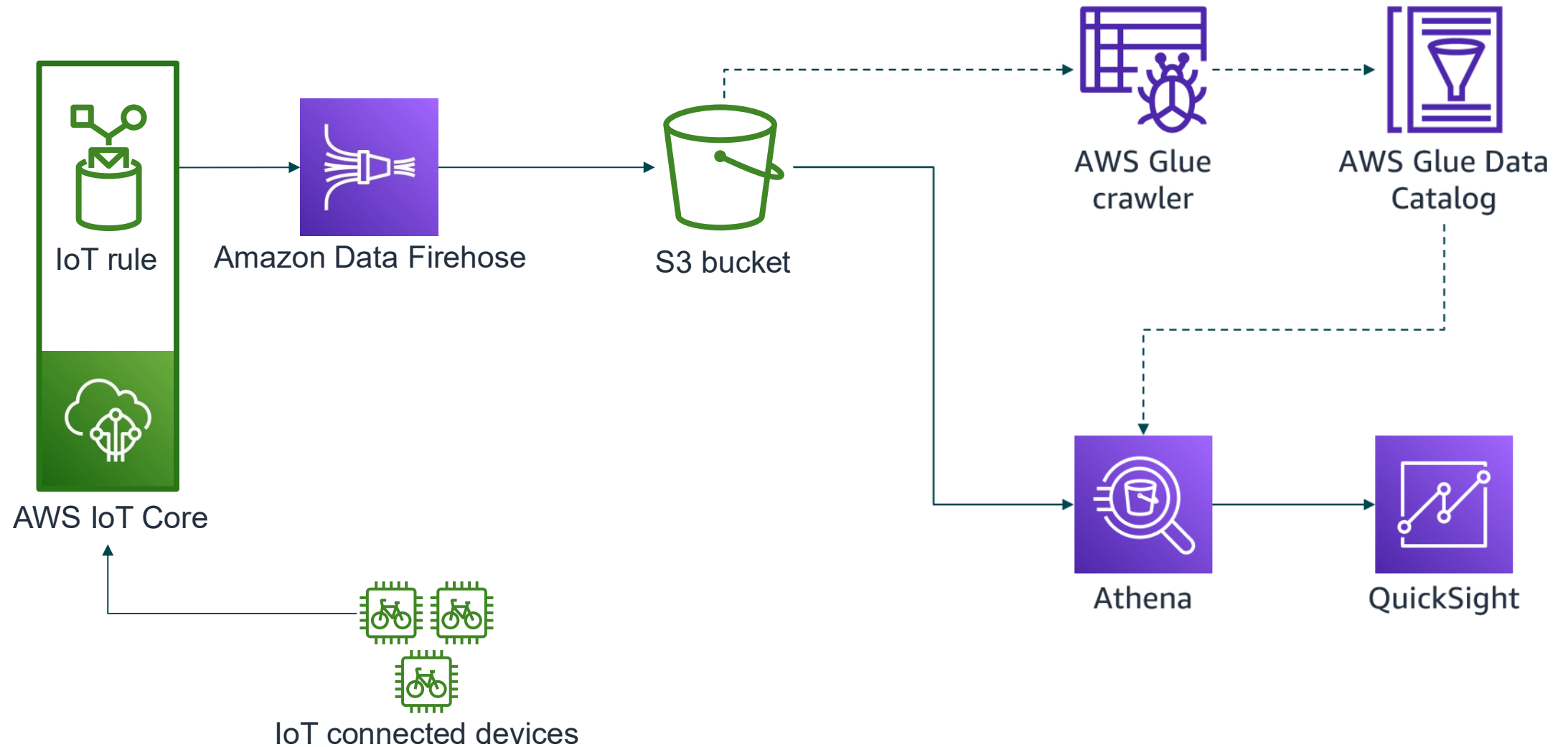
The business problem

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Working backwards from the business problem

- Your startup e-bike company has partnered with a local government to pilot an e-bike rental program in several suburban neighborhoods.
- The government's goal is to reduce carbon emissions and traffic congestion.
- At this stage, they need to collect and analyze trends in data about how the bikes are being used. They also need to determine what data they want to track as they expand the program.
- They know that they want to look at where the bikes have traveled, battery life, and engine use levels (off, low, medium, high).
- Each of your 50 bikes is equipped with an Internet of Things (IoT) device, which is powered by the bike battery and uses the wireless phone network to transmit data. Devices use the MQTT message protocol to send and receive JSON data. The devices are currently configured to send a message every 2 minutes.
- You need to build a proof-of-concept pipeline to collect and analyze data from the IoT devices that are built in to the e-bikes.

The IoT reference architecture

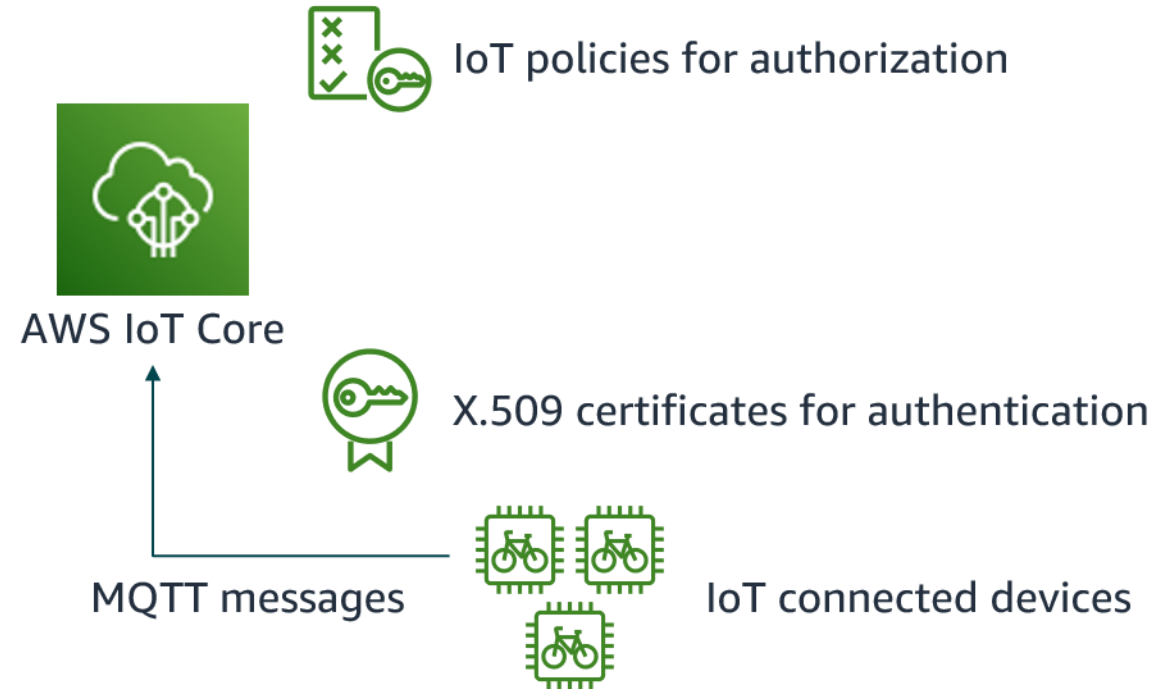


Ingestion and processing

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Connect devices to AWS IoT Core

- 1 Connect IoT devices to AWS IoT Core, and verify that messages are being published to your topic



Example: MQTT message

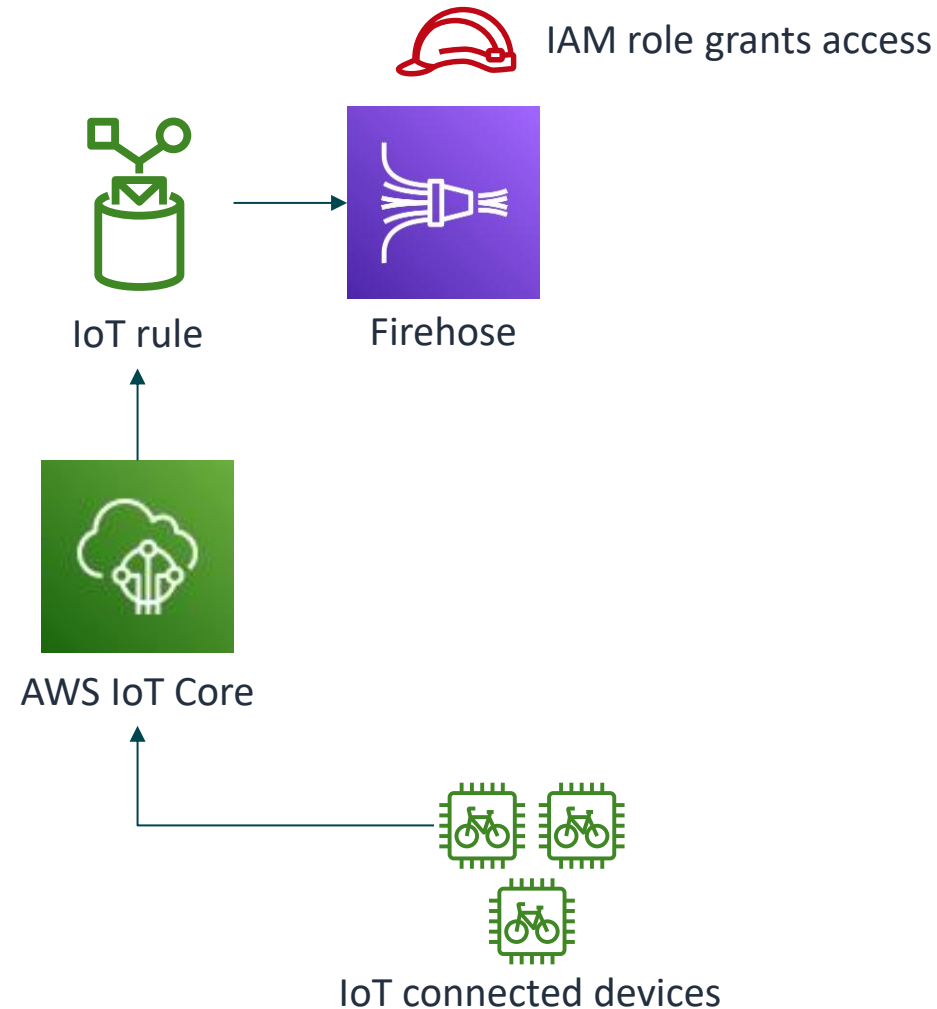
```
{  
  "format": "json",  
  "topic": "ebike/stats",  
  "timestamp": 1665102292810,  
  "payload": {  
    "device-id": "rLdMw4VRZ",  
    "location": "{ 'latitude': 38.9696, 'longitude': -77.3861 }",  
    "battery-pcnt": 75,  
    "engine-use": "off",  
    "rented": 1,  
    "last-pickup": "Oak",  
    "last-drop": "Oak"  
  }  
}
```

Topic and timestamp

Device data
collected from
sensors

Configure IoT rules to route and filter messages

- 2 Configure an IoT rule to route new messages to the Amazon Data Firehose delivery stream



Example: IoT rule

```
{
  "ruleArn": "arn:aws:iot:us-east-2:623616953939:rule/ebike",
  "rule": {
    "ruleName": "ebike",
    "sql": "SELECT device-id as id, location as bike_loc, battery-pcnt as
           battery, engine-use as engine FROM ebike/stats WHERE rented=1",
    "description": "",
    "createdAt": "2022-10-06T21:51:00+00:00",
    "actions": [
      {
        "firehose": {
          "roleArn": "arn:aws:iam::account-id:role/service-role/ebikes-role",
          "deliveryStreamName": "PUT-S3-MXjPG",
          "separator": ",",
          "batchMode": false
        }
      }
    ],
    "ruleDisabled": false,
    "awsIotSqlVersion": "2016-03-23"
  }
}
```

Filter attributes
and records

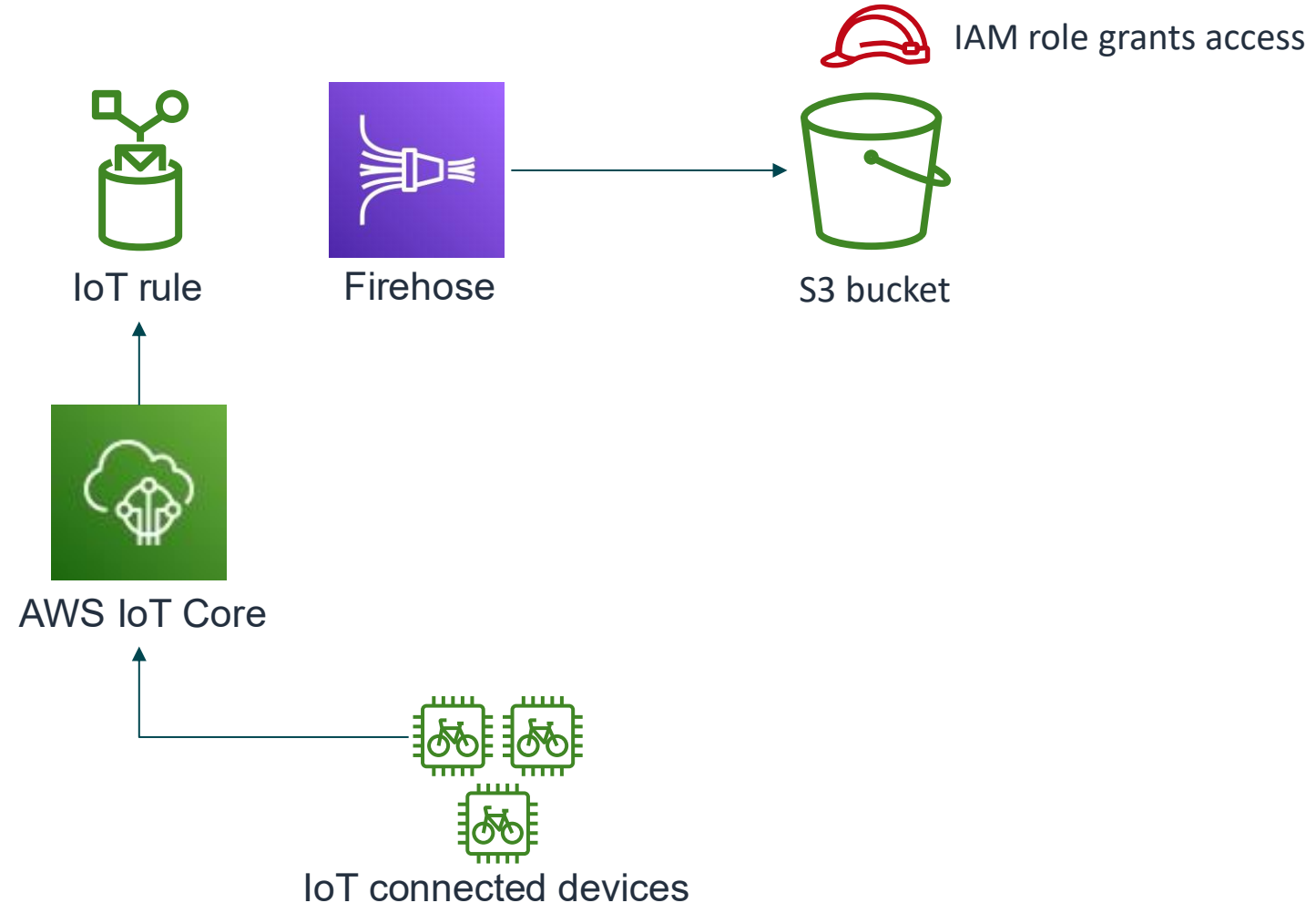
Deliver to
Firehose

IAM role that grants
permissions to put
records on the stream

...

Configure the delivery stream

- 3 Configure the Firehose delivery stream to aggregate and transform records before delivering them to Amazon S3



Configuration options for Firehose

```
{
  "DeliveryStreamDescription": {
    "DeliveryStreamName": "PUT-S3-MXjPG",
    "DeliveryStreamARN": "arn:aws:firehose:us-east-2:account-id:deliverystream/PUT-S3-MXjPG",
    "DeliveryStreamStatus": "ACTIVE",
    ...
    "DeliveryStreamType": "DirectPut",
    ...
    "Destinations": [
      {
        "DestinationId": "destinationId-0000000000001",
        "S3DestinationDescription": {
          "RoleARN": "arn:aws:iam::account-id:role/service-
arn:aws:iam::account-id:role/service-
role/KinesisFirehoseServiceRole-PUT-S3-MXjPG-us-east-2-1665071731795"
          "BucketARN": "arn:aws:s3:::ebike-stats",
          ...
          "BufferingHints": {
            "SizeInMBS": 5,
            "IntervalInSeconds": 300
          },
          "CompressionFormat": "Snappy"
        }
      }
    ]
  }
}
```

Producer puts records directly on the delivery stream

Put incoming records into the designated bucket

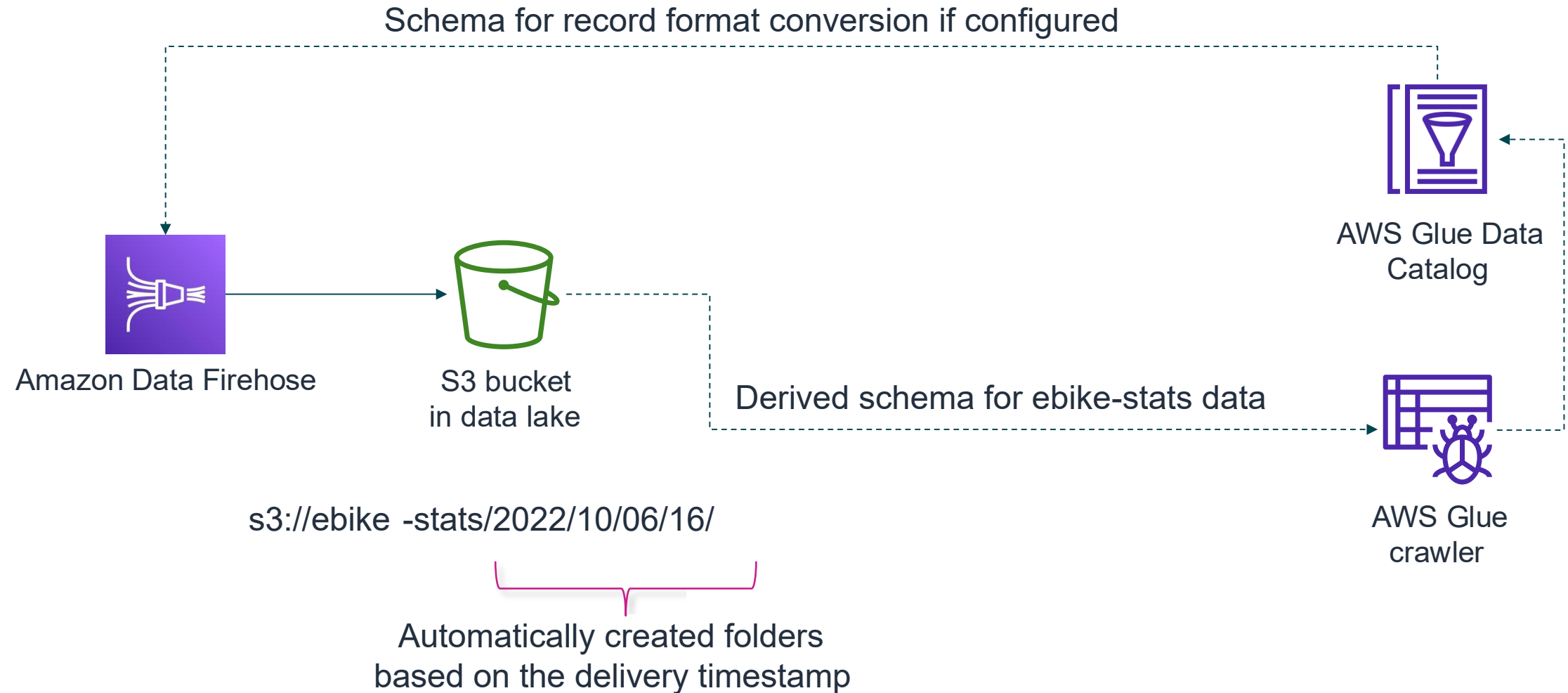
IAM role that grants permissions to put records in the bucket

Aggregate and compress records

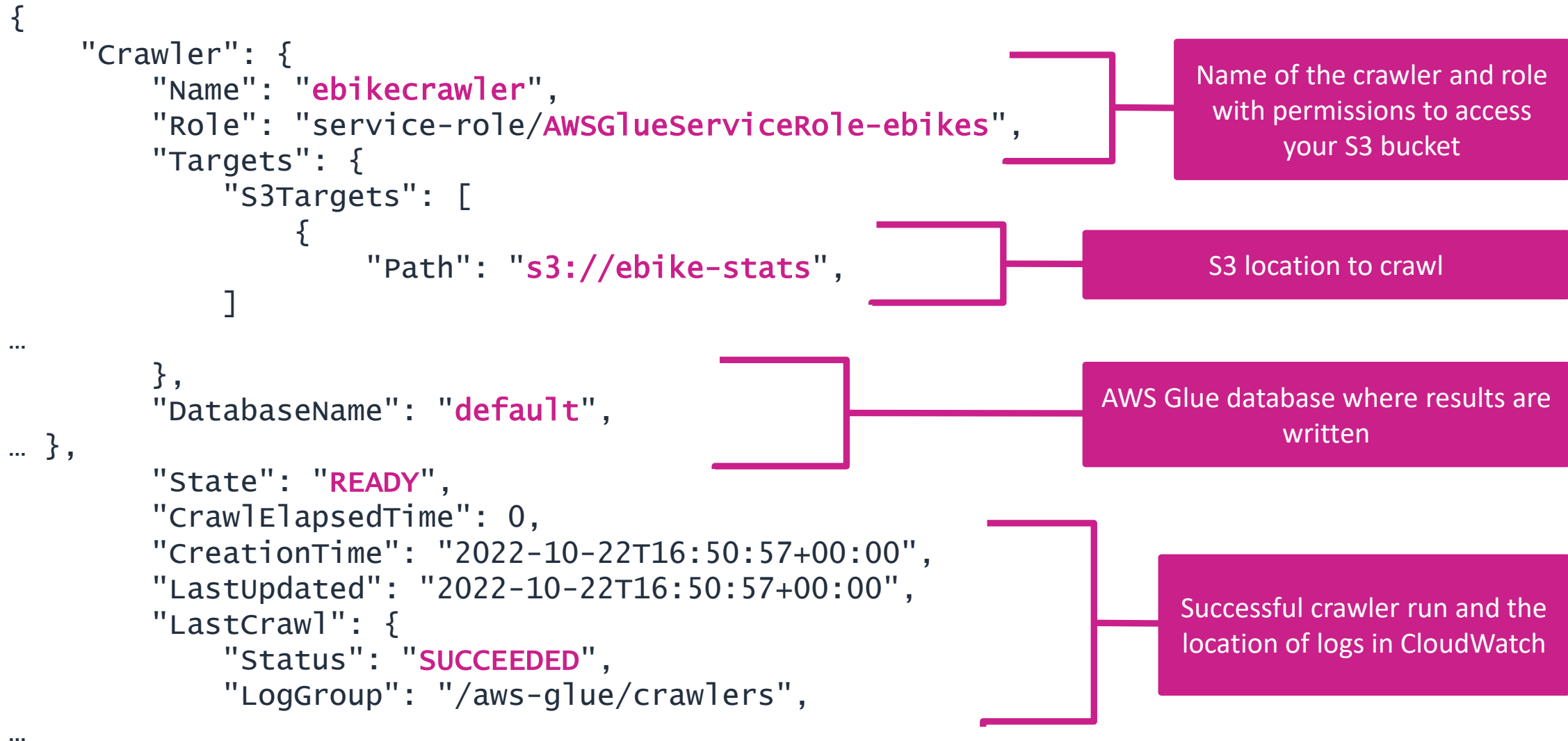
Storage

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Organize and catalog data



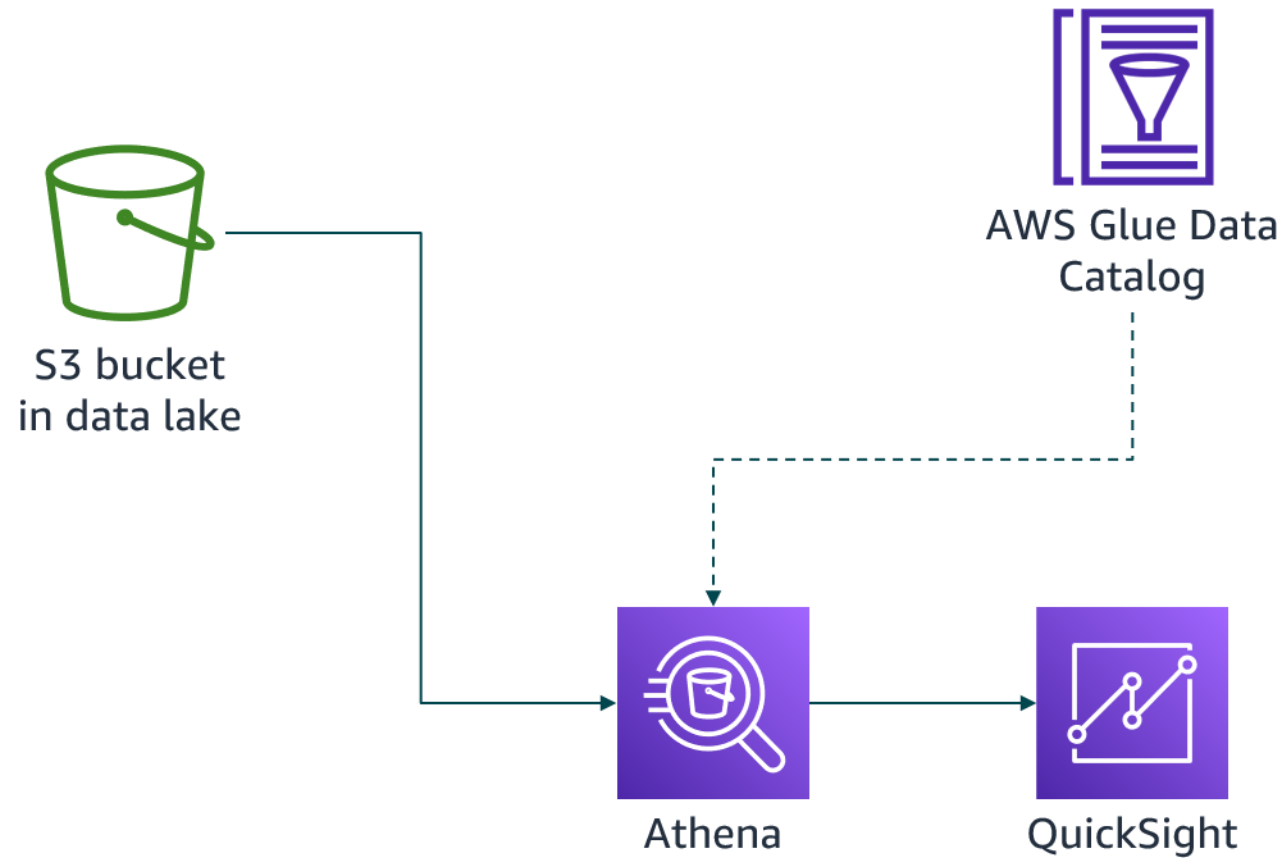
Example: Information for an AWS Glue crawler



Analysis and visualization

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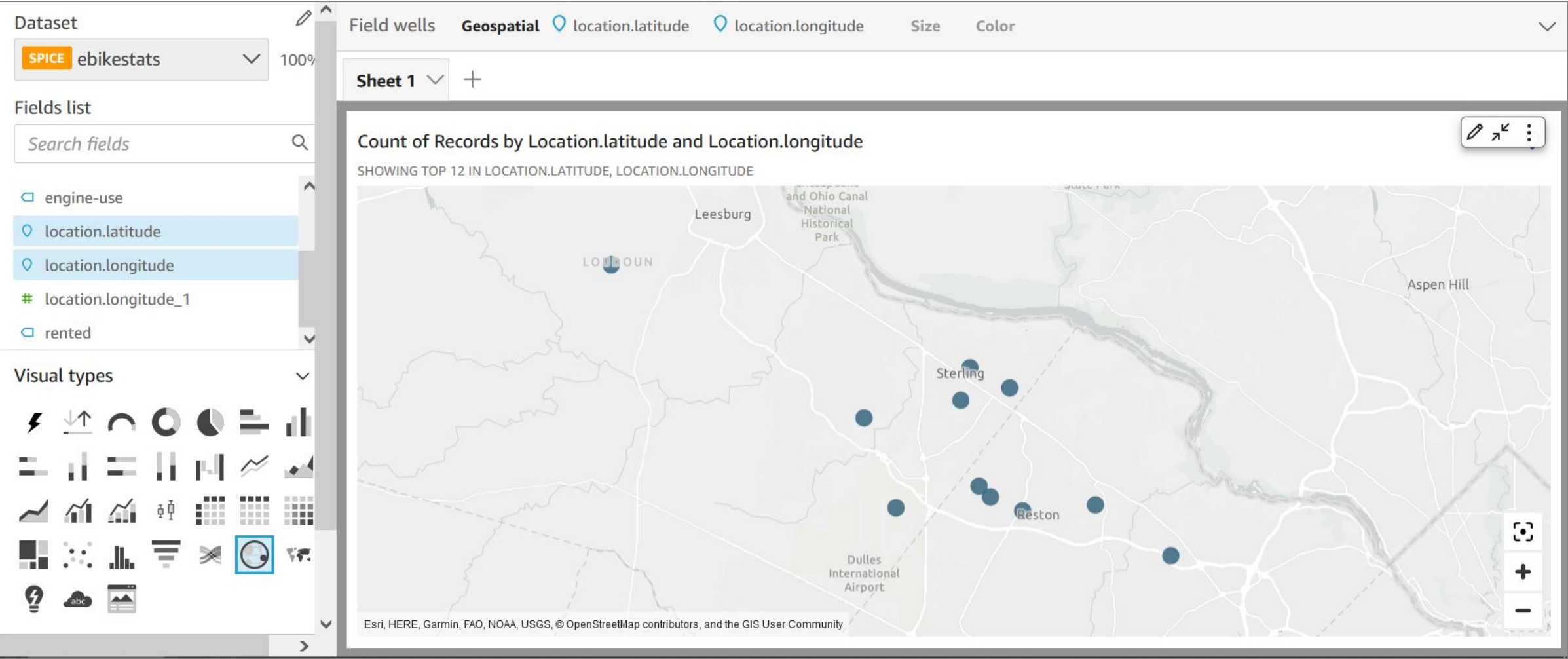
Query and visualize data in Amazon S3



Example: Athena query

```
{
  "NamedQuery": {
    "Name": "rented_bikes",
    "Database": "default",
    "QueryString": "SELECT * FROM \"default\".\"ebike_stats\" where\n\"column1.rented\" = true;",
    "NamedQueryId": "ccb0fb5c-827f-44e7-8af8-1b5ab5a8bfed",
    "workGroup": "primary"
  }
}
```

Example: Geospatial visualization in QuickSight



Wrap-up

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Review of the IoT reference architecture

